

QUEENSTOWN, AB, Canada

WMO: 710730

Lat: 50.5978N

Lon: 112.8033W

Elev: 941

StdP: 90.52

Time Zone: -7.00 (NAM)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.4	-25.5	-32.0	0.2	-28.3	-28.9	0.3	-25.3	13.2	-2.2	12.2	-2.8	2.6	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	30.7	15.7	28.7	15.1	26.7	14.5	17.9	26.0	16.8	25.0	15.9	24.4	4.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.1	12.0	21.2	13.7	10.9	20.4	12.4	10.1	18.9	54.6	26.1	50.6	24.8	47.7	24.6	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		DB	WB
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	10.8	9.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-7.5	-9.1	-2.9	4.0	10.7	14.6	17.8	16.8	12.6	4.7	-1.5	-8.6	(6)	
	DBStd	11.47	7.81	8.80	8.24	5.44	4.86	3.26	2.96	3.55	5.28	5.43	6.70	8.12	(7)	
	HDD10.0	2854	542	534	400	189	50	2	0	2	32	177	348	577	(8)	
	HDD18.3	5166	801	767	658	430	239	121	46	70	180	424	596	835	(9)	
	CDD10.0	799	0	0	0	9	72	139	241	213	112	12	2	0	(10)	
	CDD18.3	72	0	0	0	0	3	7	28	24	9	0	0	0	(11)	
	CDH23.3	1446	0	0	0	9	98	136	516	481	202	4	0	0	(12)	
	CDH26.7	444	0	0	0	1	18	32	163	167	62	1	0	0	(13)	
(14) Wind	WSAvg	4.1	4.0	3.9	4.3	4.6	4.7	4.4	3.7	3.5	4.1	4.1	4.4	3.9	(14)	
(15) Precipitation	PrecAvg	350	12	11	17	30	53	71	43	39	29	17	16	12	(15)	
	PrecMax	534	26	24	57	68	92	201	102	94	110	47	38	28	(16)	
	PrecMin	159	1	1	1	8	13	23	7	1	2	3	1	1	(17)	
	PrecStd	67	6	6	11	15	22	38	26	22	23	11	10	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	10.7	15.3	24.4	29.0	30.3	33.1	33.7	31.5	22.8	18.0	9.4	(19)	
		MCWB	3.0	4.2	6.3	10.4	12.7	16.1	16.4	15.6	14.4	11.6	8.1	2.2	(20)	
	2%	DB	5.3	6.1	13.0	19.6	26.2	26.9	30.7	30.6	28.6	19.1	12.6	5.6	(21)	
		MCWB	1.6	1.7	5.3	8.5	12.0	14.1	16.5	15.6	13.8	9.4	5.8	1.4	(22)	
	5%	DB	3.8	4.2	10.5	16.1	23.4	24.5	28.3	28.4	25.6	16.2	9.7	3.7	(23)	
		MCWB	0.9	0.6	4.1	6.5	11.2	14.0	16.1	15.1	12.9	8.1	4.2	0.4	(24)	
	10%	DB	2.6	1.8	7.3	13.2	20.5	22.5	26.2	25.9	22.5	13.3	7.3	2.2	(25)	
		MCWB	0.2	-0.8	2.7	5.1	9.8	13.1	16.0	14.5	11.8	6.4	2.5	-0.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	4.8	7.3	11.0	13.3	18.1	20.5	18.2	15.7	12.2	8.1	3.1	(27)	
		MCDB	7.1	10.6	13.8	22.5	26.1	25.8	27.1	27.2	27.5	21.3	16.8	7.4	(28)	
	2%	WB	1.9	2.1	5.7	8.8	12.3	16.1	18.6	16.9	14.6	10.2	6.1	1.8	(29)	
		MCDB	4.8	6.1	12.0	17.9	24.2	23.9	25.6	26.6	25.7	17.4	11.9	5.4	(30)	
	5%	WB	1.1	0.6	4.3	7.0	11.4	15.0	17.5	16.0	13.3	8.5	4.5	0.7	(31)	
		MCDB	3.7	3.8	9.8	14.9	22.0	22.1	25.3	25.8	24.1	15.1	9.3	3.6	(32)	
	10%	WB	0.2	-0.7	2.6	5.6	10.4	13.8	16.5	15.1	12.2	7.0	2.9	-0.7	(33)	
		MCDB	2.7	1.6	7.1	12.3	19.1	21.0	24.3	24.2	21.4	12.7	6.9	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.1	10.9	12.6	13.5	12.9	15.0	15.6	14.2	12.5	10.5	9.1	(35)	
		MCDBR	9.7	12.2	14.5	17.8	18.1	16.0	18.6	19.8	18.8	16.8	13.5	10.9	(36)	
	5% WB	MCWBR	7.1	8.5	8.6	8.8	7.4	6.9	7.7	7.6	7.4	8.6	7.8	7.9	(37)	
		MCDWR	9.8	11.5	13.8	16.9	16.3	14.0	15.5	17.2	17.7	16.0	13.3	10.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.257	0.264	0.284	0.333	0.352	0.345	0.346	0.362	0.320	0.281	0.254	0.236	(40)	
		taud	2.418	2.474	2.475	2.341	2.340	2.406	2.424	2.358	2.453	2.537	2.498	2.435	(41)	
	Edn at Noon	Edn at Noon	802	896	932	909	903	911	904	869	877	862	806	780	(42)	
		Edh at Noon	59	73	87	113	118	112	108	110	89	68	54	49	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.21	2.19	3.52	4.82	5.76	6.30	6.84	5.47	3.87	2.38	1.30	0.90	(44)	
	RadStd	RadStd	0.09	0.21	0.26	0.34	0.41	0.40	0.34	0.36	0.36	0.27	0.12	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page